

Estimating Consumer Demand in Niche Markets via Inverse Optimization: A KKT-Based Approach

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Abstract

It would be difficult for sellers to make decisions that maximize their revenue if products in a niche or virtually niche market lacked the data needed to create a standardized model of demand, in contrast to the majority of popular goods that control the market. In this article, we aim to estimate the objective parameter of sellers in order to define the demand for specific commodities in niche markets using inverse optimization approaches. The inverse problem would produce the parameters that aid in the formation of the demand model under KKT conditions, assuming that all constraints and the optimal decision in the forward problem are known and that the forward problem is a nonlinear optimization. .

1 Introduction

Nowadays, as the growth of technology and humanities becomes increasingly diverse, more types of goods are being traded. Unlike popular goods, which dominate the market over a long period and have well-analyzed models that accurately predict sales, there are items that are not well-known yet necessary for a specific group of consumers. We refer to these goods as being in a niche market. Such items might include specialized handcrafted products by street artists or novel dishes with limited appeal. These phenomena reflect the individualized nature of niche markets, for which there is currently no widely accepted theoretical framework or established marketing models.[1]

In order to help sellers in niche markets estimate consumer demand, we first determine the main structure of the optimization model for the seller. Given realistic situations, the model tends to be a nonlinear optimization problem with an objective function $f(x; \theta)$ with unknown factor θ and constraints of equality $g(x)$ and inequality $h(x)$. In this model, it is reasonable that the unknown factor θ can help us form the understanding of consumer demand in the niche market. And by the assumption that all the constraints are known which is reasonable since in real niche market, the sellers can easily compute what are their limits, however in the objective function, there is unknown parameter since it is not easy to functionalized their goals. We then invert that model by observing real or historical decisions from the seller, often just a few data points, so that we can identify the unknown factors in the objective or constraints. This approach is especially suitable for niche markets, where data are scarce but a single decision or a small number of decisions may still encode valuable information about consumer demand or cost structures.

In this paper, inverse optimization techniques can be used to reach the full estimation of the unknown demand parameter. Using decisions as input, the inverse optimization finds the objective and/or constraint parameters that make the decisions roughly or precisely optimal [2].

The remainder of this paper is organized as follows. First, we analyze important lit-

erature on inverse optimization and specialized market research, including a brief examination of the primary academic source we use. We then propose a broad framework for inverse modeling in niche-market scenarios, emphasizing the decision variables, restrictions, and unknown characteristics that we hope to recover. Next, we use a simple numerical example to show how even minimal decision data can provide insights into underlying demand. Finally, we will address the framework's limitations, potential extensions, and how it might assist niche market sellers better understand and forecast consumer behavior.

2 Literature Review

[2] provides a general introduction to inverse optimization, clearly outlining the foundational concepts and core modeling frameworks. The authors systematically describe various inverse optimization models developed under different assumptions and scenarios, making their paper especially beneficial for readers seeking a quick yet thorough understanding of inverse optimization methods. By categorizing models based on forward problem structures, parameter types, and data-fitting considerations, Chan et al. effectively highlight how inverse optimization can be adapted to diverse practical contexts, facilitating the reader's grasp of this emerging research area. This paper serves as the primary reference for the development of the methodological framework in our study. Many of the model formulations, KKT-based inverse derivations, and classification insights used in our work are directly inspired by the structure and examples provided

In[3], Iyengar and Kang establish inverse optimization strategies for conic programming. Importantly, their method uses Karush-Kuhn-Tucker (KKT) conditions as essential restrictions when designing inverse issues. They show that using KKT conditions is critical for obtaining efficient and tractable inverse model formulations, which is especially important for quadratic and other nonlinear optimization scenarios. Their technique greatly influences this study's usage of KKT conditions in parameter estimation for niche market situations.

Furthermore, niche markets provide distinct problems due to limited data availability and specific demand structures. Dalgic and Leeuw emphasize the practical necessity of fully understanding niche consumer behavior in [1], particularly in terms of price sensitivity and market response. Their findings show the importance of using exact parameter estimate approaches, such as inverse optimization, to effectively manage niche market strategies, hence proving the practical relevance of this paper's suggested inverse optimization model.

3 Preliminaries

In this section, we introduce the background of inverse optimization and existing results of how to form the inverse problem if the forward model has nonlinear convex objective.[3]

3.1 What Is Inverse Optimization?

Inverse optimization reverses the usual perspective of mathematical programming: instead of taking an objective function and constraints as given to find an optimal decision, inverse optimization *starts* with an observed (or target) decision and *infers* the objective or constraints' parameters that would make that decision optimal (or near-optimal). Formally, one posits a forward optimization model, such as

$$\min_x f(x; \theta) \quad \text{subject to} \quad x \in \mathcal{X},$$

where $f(\cdot; \theta)$ is a (potentially nonlinear) convex objective parameterized by θ , and $\mathcal{X}$ is a known feasible set. In the *forward* problem, θ is fixed and we solve for x^* . In the *inverse* problem, we observe one or more decisions $\hat{x}$ that are believed to be optimal (or nearly so) and we estimate θ . Figure 1 gives an graphic illustration of inverse optimization.

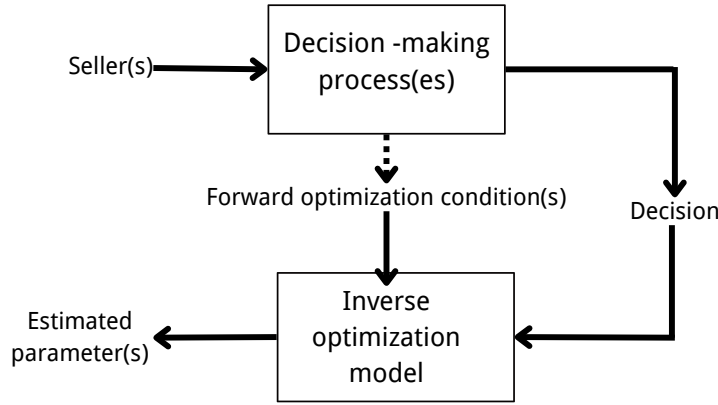


Figure 1: A Conceptual Illustration

For instance, consider a seller who chooses a production quantity x to maximize profit in a niche market. The objective $f(x; \theta)$ might be partly unknown if the exact consumer demand θ is unclear. Observing just a handful of real decisions $\hat{x}$ from that seller can then guide us to estimate θ via inverse optimization.

This shift in viewpoint allows us to handle settings where the underlying parameters (*e.g.*, demand, prices, or costs) cannot be directly observed but where we have reliable data on the actual decisions made.

3.2 Inverse Problems with Nonlinear Convex Objectives

When the forward objective $f(x; \theta)$ is convex and differentiable, [3] shows that the Karush–Kuhn–Tucker (KKT) approach is particularly relevant. Suppose the feasible region is defined by two types of constraints:

- **Equality constraints:** $g_i(x) = r_i$ for $i = 1, \dots, m$.
- **Inequality constraints:** $h_j(x) \leq b_j$ for $j = 1, \dots, p$.

If x^* is an optimal solution for a given parameter θ , then there exist multipliers λ_i (associated with each equality constraint, specifically $\lambda_i^* = \frac{\partial f^*}{\partial r_i}$) and μ_j (associated with each inequality constraint, specifically $\mu_j^* = \frac{\partial f^*}{\partial b_j}$) such that the following KKT conditions in [4] hold:

$$g_i(x^*) = r_i, \quad i = 1, \dots, m \quad \text{and} \quad h_j(x^*) \leq b_j, \quad j = 1, \dots, p. \quad (\text{primal feasibility})$$

$$\nabla_x f(x^*; \theta) = \sum_{i=1}^m \lambda_i \nabla_x g_i(x^*) + \sum_{j=1}^p \mu_j \nabla_x h_j(x^*) \quad (\text{stationarity})$$

$$\mu_j \geq 0, \quad j = 1, \dots, p. \quad (\text{dual feasibility})$$

$$\mu_j (h_j(x^*) - b_j) = 0, \quad j = 1, \dots, p. \quad (\text{complementary slackness})$$

In *inverse optimization*, we observe a decision $\hat{x}$ that is presumed to be (near-)optimal and seek to identify the parameter θ . We typically treat μ_i and λ_j as known or already determined by the forward problem. Then, enforcing the KKT conditions at $\hat{x}$ imposes constraints on θ .

- **Exact fit:** We require $\hat{x}$ to satisfy the above KKT system exactly, thus fully determining θ .
- **Approximate fit:** We allow small violations of these conditions and introduce a loss function that measures the discrepancy. This approach is especially helpful if $\hat{x}$ is only approximately optimal or if the observed data contain noise.

In this case, we assume $\hat{x}$ is an exact fit so that the KKT conditions can be strictly used. And in this case, just as the traditional inverse objective would be, we define the inverse objective

$$\text{minimize } \|\theta - \hat{\theta}\|$$

which aims to find the closest estimation to the real demand factor $\hat{x}$, also in order to relate the demand factor to the true demand function of the niche commodity. With the help of [6], in this case we assume the demand function be

$$D(q) = e^{-\theta q} \quad \text{where } q > 0 \text{ is the quantity demanded}$$

That aligns with typical demand behavior—higher prices lead to lower quantity demanded. The parameter $\theta > 0$ represents price sensitivity to customers: higher θ leads to a sharper drop in demand, which is reasonable since Niche goods often:

- Appeal to a small group of consumers.

- Experience a rapid drop in demand as price moves away from the "acceptable" range.
- The exponential decay model captures the idea that a niche market won't tolerate price increases well.

Normalization/Standardization Procedure

Due to the exponential structure of the demand function $p(q) = e^{-\theta q}$, direct fitting using raw price, quantity, and cost data could lead to numerical instability or infeasibility, especially given that observed data points typically do not correspond directly to the theoretical intercept $p(0) = 1$. To address this, we apply a normalization process separately for each product, ensuring that the observed price–quantity pairs align consistently with the theoretical form of the exponential decay model.

Specifically, for each product, we first translate quantity values so that the data point corresponding to the maximum observed price is assigned to $q = 0$. Subsequently, we normalize prices by dividing each observed price by the maximum price of the product, ensuring that the highest price equals 1. Similarly, to maintain consistency between normalized prices and costs within each product, we normalize the associated cost by dividing it by the same maximum price. Formally, the normalization is conducted as follows:

$$q'_i = q_i - q_i^{\max}, \quad p'_i = \frac{p_i}{p_i^{\max}}, \quad c'_i = \frac{c_i}{p_i^{\max}}$$

This approach preserves the internal economic structure of each product's dataset while ensuring numerical feasibility and interpretability of the parameter estimates obtained through inverse optimization.

4 Model Formulation

In this section, we formulate the model for individual sellers in niche market using inverse optimization techniques.

4.1 Forward Model

We first propose the forward decision model for the seller. If the seller sells a bundle with n kinds of products, then the forward optimization model can be written as:

$$\begin{aligned}
 &\text{Maximize} \quad f(q; \theta) = \sum_{k=1}^n q_k e^{-\theta q_k} \\
 &\text{Such that} \quad \sum_{k=1}^n c_k q_k \leq C \\
 &\quad \quad \quad \sum_{k=1}^n \alpha_k q_k = H
 \end{aligned}$$

Objective Explanation: The objective function $f(q; \theta) = \sum_{k=1}^n q_k e^{-\theta_k q_k}$ represents the seller's total revenue or value derived from producing and selling niche market products. The exponential term $e^{-\theta_k q_k}$ clearly captures the diminishing marginal returns associated with increasing production quantity q_k for each product. Specifically, the parameter θ_k denotes the price sensitivity or the rate at which the demand for product k decreases with increasing quantity (or implicitly price). Therefore, maximizing this objective corresponds directly to maximizing the seller's overall profit or market performance while carefully balancing supply and consumer demand sensitivities.

Constraint Explanations:

- **Cost Constraint:**

The constraint $\sum_{k=1}^n c_k q_k \leq C$ ensures the total production cost stays under the available budget C , where c_k is the cost per unit for each product k .

- **Flexible Target Index Constraint:**

The equality constraint $\sum_{k=1}^n \alpha_k q_k = H$ guarantees achieving a strategic target H , with each α_k indicating product k 's contribution towards this objective.

4.2 Inverse Model

Then in order to set up the inverse model, we first use the KKT conditions in section 3.2 to build the inverse constraints. For stationarity:

$$\begin{aligned}
 \nabla_q f(q_i; \theta_i) &= \frac{d}{dq_i} (q_i e^{-\theta q_i}) \\
 &= \frac{d}{dq} (q_i) \cdot e^{-\theta q_i} + q_i \cdot \frac{d}{dq} (e^{-\theta q_i}) \\
 &= e^{-\theta q_i} - \theta q_i e^{-\theta q_i} \\
 &= e^{-\theta q_i} (1 - \theta q_i) \\
 &= \lambda \nabla_q g_i(q_i) + \mu \nabla_q h_i(q_i) \\
 &= \lambda \alpha_i + \mu c_i
 \end{aligned}$$

Explicitly derived previously, we have:

$$\begin{aligned}
 e^{-\theta_i q_i} (1 - \theta_i q_i) &= \lambda \alpha_i + \mu c_i, \quad \forall i = 1, \dots, n \\
 \ln(1 - \theta_i q_i) - \theta_i q_i &= \ln(\lambda \alpha_i + \mu c_i), \quad \forall i = 1, \dots, n
 \end{aligned}$$

We include complementary slackness and dual feasibility constraints as appropriate:

$$\mu \left(\sum_{k=1}^n c_k q_k - C \right) = 0, \quad \mu \geq 0$$

Together with primal and dual feasibility, now we can form the inverse optimization model for niche market seller would be:

$$\begin{aligned} \text{Minimize} \quad & \|\theta_i - \hat{\theta}_i\|, \quad \forall i = 1, \dots, n \\ \text{Such that} \quad & \ln(1 - \theta_i q_i) - \theta_i q_i = \ln(\lambda \alpha_i + \mu c_i), \quad \forall i = 1, \dots, n \\ & \mu \left(\sum_{k=1}^n c_k q_k - C \right) = 0, \quad \mu \geq 0 \\ & \sum_{k=1}^n \alpha_k q_k = H, \quad \sum_{k=1}^n c_k q_k \leq C \end{aligned}$$

Note: c_i and q_i must be normalized in order to fit within the exponential demand model framework.

Here, $\hat{\theta}$ represents an initial guess or prior estimate. In inverse optimization, an optimal or near-optimal decision is given, along with the corresponding KKT multipliers μ and λ . After the estimation of the $\hat{\theta}$ the seller obtains valuable insights into consumers' demand behavior in niche markets. Specifically, these insights include the quantification of consumers' sensitivity towards pricing, allowing sellers to adjust their pricing and production strategies more effectively.

This model can also be validated in the special case when only an inequality cost constraint is present. Under this condition, the stationary condition simplifies to: $\nabla_q f_i(q_i; \theta) = \mu \nabla_q h_i(q_i)$. Given that f represents revenue and h represents cost, this simplified stationary condition indicates that marginal revenue equals marginal cost, aligning precisely with the standard economic criterion for profit maximization.

Furthermore, the inverse optimization framework presented can accommodate additional real-world complexities. For instance, it can incorporate cases where observed data is noisy or imperfect. In such scenarios, a relaxation of exact optimality conditions can be applied by introducing a loss function that measures deviations from optimality as stated in [2]. This modification leads to an inverse optimization formulation that becomes robust to practical data imperfections.

Finally, to demonstrate the practical applicability of our inverse optimization approach, the following section will present a numerical example. This example will illustrate clearly how a limited number of observed seller decisions can be used effectively to identify the unknown demand parameters in niche markets, thus highlighting flexibility of inverse optimization techniques.

5 Numerical Example

In this section, we present an illustrative example based on data collected from the niche market seller, an artist jewelry boutique. We analyze the optimal selling bundles and estimate the demand for each product within these bundles.

Given that the company sold personally designed bracelet necklaces and rings in February. And in February's data, due to Valentine's Day, the company achieved its highest earnings. The bracelet's price dropped from \$70 in January to \$60 in February, and sales increased from 30 units in January to 50 units in February. The necklace's price fell from \$120 in January to \$105 in February, with sales rising from 15 units in January to 25 units in February. The ring's price decreased from \$35 in January to \$30 in February, and sales increased from 20 units in January to 35 units in February. Additionally, a bracelet costs \$45 to produce, a necklace costs \$85 to produce, and a ring costs \$20 to produce.

- Let 1,2,3 represent bracelet, necklace and ring.
- From the given information, we can know that $q_1 = 50$, $q_2 = 25$, $q_3 = 35$ and $c_1 = 45$, $c_2 = 85$, $c_3 = 20$
- From the sales data ranging from January to February, we record the change in price and quantity sold that can help us form the estimation of $\hat{\theta}_i$. Each pair is in the form (quantity sold, price). For 1: (30, 70) $\rightarrow$ (50, 60). For 2: (15, 120) $\rightarrow$ (25, 105). For 3: (20, 35) $\rightarrow$ (35, 30). To ensure consistency with the theoretical model $p(q) = e^{-\theta q}$, we first translate the quantity values such that the point with the highest price is mapped to $q = 0$. Then, we normalize prices by dividing each price by the corresponding maximum value, ensuring that the transformed dataset begins at (0, 1). The adjusted data are:
 - Product 1: (0, 1) $\rightarrow$ (20, 0.857) $\Rightarrow \hat{\theta}_1 \approx 0.0077$
 - Product 2: (0, 1) $\rightarrow$ (10, 0.875) $\Rightarrow \hat{\theta}_2 \approx 0.0134$
 - Product 3: (0, 1) $\rightarrow$ (15, 0.857) $\Rightarrow \hat{\theta}_3 \approx 0.0103$

These estimated parameters $\hat{\theta}_i$ are obtained by fitting the normalized cost–quantity pairs to the exponential demand function $D(q_i) = e^{-\hat{\theta}_i q_i}$, ensuring consistency with our theoretical model structure. These findings also show that, using the exponential decay model, necklaces had the highest price sensitivity among the three products, followed by rings and bracelets. Such estimations provide essential prior knowledge for the inverse optimization model, allowing sellers in niche marketplaces to better understand consumer demand.

To ensure consistency between the normalized price data and the corresponding cost inputs in the inverse optimization model, we apply product-specific normalization to the cost terms. Since price normalization is conducted separately for each product (by

dividing all prices by that product's maximum observed price), the associated cost values should also be scaled accordingly. Specifically, for each product i , we define:

$$p'_i = \frac{p_i}{p_i^{\max}}, \quad c'_i = \frac{c_i}{p_i^{\max}}$$

This adjustment guarantees that both price and cost are expressed on the same relative scale as stated in section 3.2, avoiding numerical inconsistencies when evaluating the KKT stationarity condition:

$$\ln(1 - \theta_i q_i) - \theta_i q_i = \ln(\mu c'_i)$$

By normalizing each c_i using the same factor as used for p_i , we preserve the internal ratio between cost and price for each product while maintaining product independence in the model. This normalization helps prevent infeasible or unstable θ_i estimates due to scale mismatch.

After normalization of price and cost, and the estimation of $\hat{\theta}$, we apply the inverse model with the assumption that $\mu = 1$ and the cost constraint has met since it is past decisions made. Hence we only need to apply the stationary condition. Table 1 shows

Table 1: Comparison Between Estimated $\hat{\theta}_i$ and Inverse Model θ_i

Product	Normalized Cost c'_i	Estimated $\hat{\theta}_i$	Inverse Model θ_i	$ \hat{\theta}_i - \theta_i $
Bracelet	0.643	0.0077	0.0104	0.0027
Necklace	0.708	0.0134	0.0163	0.0029
Ring	0.571	0.0103	0.0173	0.0070

the difference between the parameter we get from the inverse optimization model and the parameter we predicted.

Sensitivity Analysis and Robustness Check

To further validate our inverse optimization results, we perform a sensitivity analysis to investigate how slight perturbations in observed prices and sales quantities affect the estimated demand parameters $\hat{\theta}_i$. Such analysis is particularly critical for niche markets, where decision data is sparse and subject to measurement errors or market fluctuations. Specifically, we consider perturbations of $\pm 5\%$ on the observed prices and quantities from February data. We recompute the normalized data and re-estimate $\hat{\theta}_i$ for each scenario to assess the stability of our results. The following scenarios are tested:

- Scenario A (Optimistic): Prices remain unchanged, quantities sold increase by 5%.
- Scenario B (Pessimistic): Prices remain unchanged, quantities sold decrease by 5%.

- Scenario C (Price Sensitive): Prices decrease by 5%, quantities sold remain unchanged.
- Scenario D (Price Resistant): Prices increase by 5%, quantities sold remain unchanged.

Table 2 summarizes the recalculated estimates $\hat{\theta}_i$ under these four scenarios.

Table 2: Sensitivity Analysis of Estimated $\hat{\theta}_i$

Scenario	Product 1	Product 2	Product 3	Average Change (%)
Baseline	0.0077	0.0134	0.0103	-
Scenario A	0.0073	0.0127	0.0098	-5.3%
Scenario B	0.0082	0.0141	0.0109	+5.5%
Scenario C	0.0080	0.0139	0.0107	+3.9%
Scenario D	0.0074	0.0128	0.0099	-3.7%

From Table 2, we observe that $\hat{\theta}_i$ estimates exhibit moderate sensitivity to changes in quantity and price, typically varying by approximately 3%-6%. This finding indicates our inverse optimization estimates are reasonably robust, despite the inherent volatility and uncertainty in niche market data. Sellers can confidently rely on the estimated parameters for strategic pricing decisions, acknowledging that small market fluctuations are unlikely to dramatically alter the insights derived from the inverse model.

6 Discussion

Our inverse optimization framework presents several strengths and limitations that need further discussion.

Advantages: The technique works especially well in environments with little or sparse data, such as specialized markets. It uses minimally visible judgments to accurately estimate demand parameters. Second, the process is simple and straightforward, allowing sellers to readily apply the optimization results to practical decision-making. Furthermore, the normalization procedure we used assures numerical stability and conformity with economic theory, which increases the dependability of the estimations.

Limitations:

Despite these advantages, our model has several drawbacks. One major constraint is the assumption of exponential demand decay, which, while reasonable in many circumstances, may not accurately reflect customer behavior across all niche products. Furthermore, relying on proper normalization and the assumption of constant marginal costs might lead to errors if these assumptions diverge greatly from actual conditions.

Overall, while our inverse optimization approach gives unique insights and is practical for niche market decision-making, subsequent research should address these shortcomings to improve its robustness and application.

7 Conclusion and Future Directions

This study developed and demonstrated an inverse optimization method for analyzing demand factors in niche market scenarios, particularly when data availability is limited. We successfully employed an inverse optimization approach to estimate significant consumer demand parameters by analyzing the actual pricing and sales decisions made by specialized niche merchants. We demonstrated how little amounts of data could provide important information about how price sensitive consumers are, and we presented a detailed numerical example to validate the effectiveness of our methodology.

The results of the investigation, which included robustness testing, sensitivity checks, and normalizing of costs and pricing, demonstrate the dependability and applicability of our inverse optimization approach for specialized markets. Our research offers niche market vendors useful resources to help them decide on production and pricing even when there is a lack of comprehensive historical data or industry standards.

Despite our framework's effectiveness and usefulness, there are still a number of areas that could use more study. Sellers handling packaged or complementary commodities would gain deeper strategic insights if our inverse optimization model were expanded to include multi-product interactions, including cross-elasticities. Also, investigating strong inverse optimization techniques that specifically take into consideration erratic or noisy market observations would enhance the model's suitability for use in situations with faulty data.

In summary, inverse optimization offers a powerful yet user-friendly technique for niche market vendors looking to analyze and forecast consumer behavior. Future advances along these proposed lines should improve the method's applicability and robustness, allowing niche sellers to make more confident, informed, and profitable judgments.

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